



London Ambulance Service



NHS Trust

Anatomy and Physiology – Part Component 2 and 5

The Cardiovascular (CV) System



Cardiovascular System - Part Component 2+5

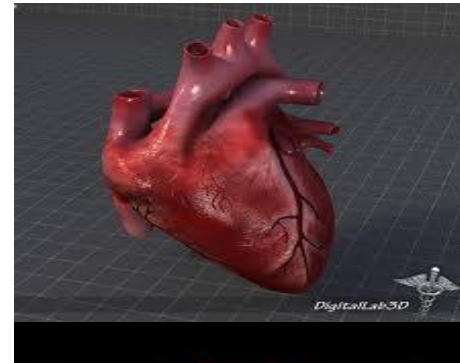
Objectives

- State functions of the cardiovascular system
- Understand the structures of the cardiovascular system
- Identify the main structures of the heart
- Identify the blood vessels
- Identify the electrical pathway of the heart

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Components of the CV System

- Heart
- Blood vessels
- Blood





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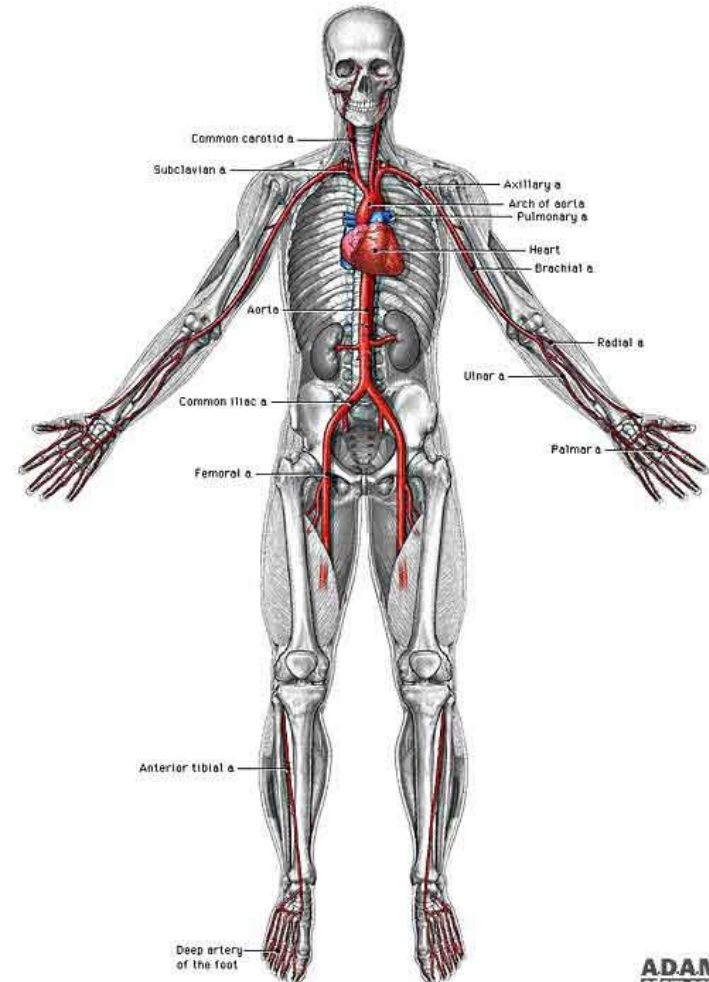
Functions of the CV System

- Composed of blood vessels
- Heart pumps blood around the body
- Transport of nutrients
- Removal of waste products
- Hormonal Control
- Temperature regulation
- Host defence
- Reproduction

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Divisions:

- Systemic
- Pulmonary
- Coronary
- Hepatic Portal



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Liquid Part – 55%

- Plasma

Solid Part - 45%

- Erythrocytes (red cells)
- Leucocytes (white cells)
- Thrombocytes (platelets for clotting)



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Blood Vessels



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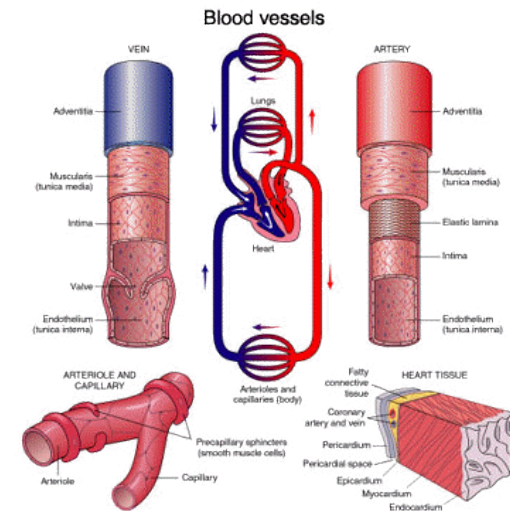
Blood Vessels

There are five types of blood vessel:

Arteries - **Arterioles** - **Capillaries** - **Venules** - **Veins**

All blood vessels have three layers:

- An outer fibrous sheath (Tunica Adventitia)
- A middle layer of muscle and elastic fibre (Tunica Media)
- A smooth inner lining of endothelium (Tunica Intima)

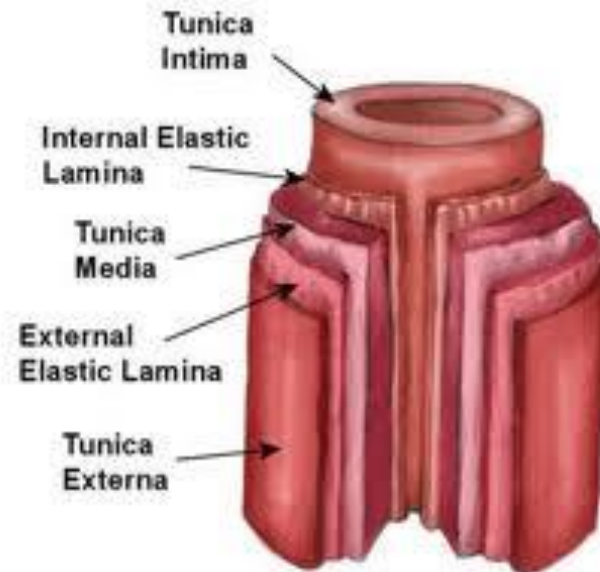


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Blood Vessels

Arteries - Arterioles

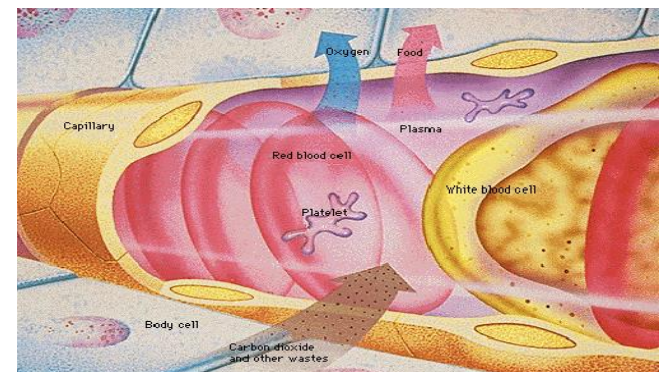
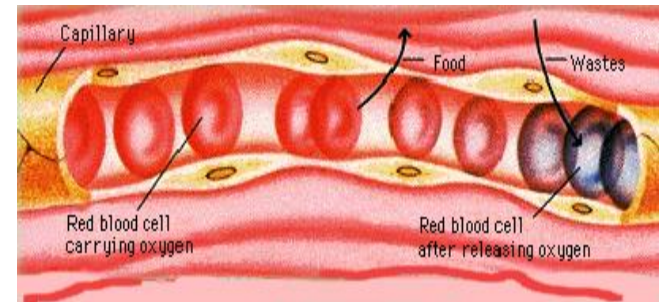
- Arteries carry blood away from the heart



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Capillaries

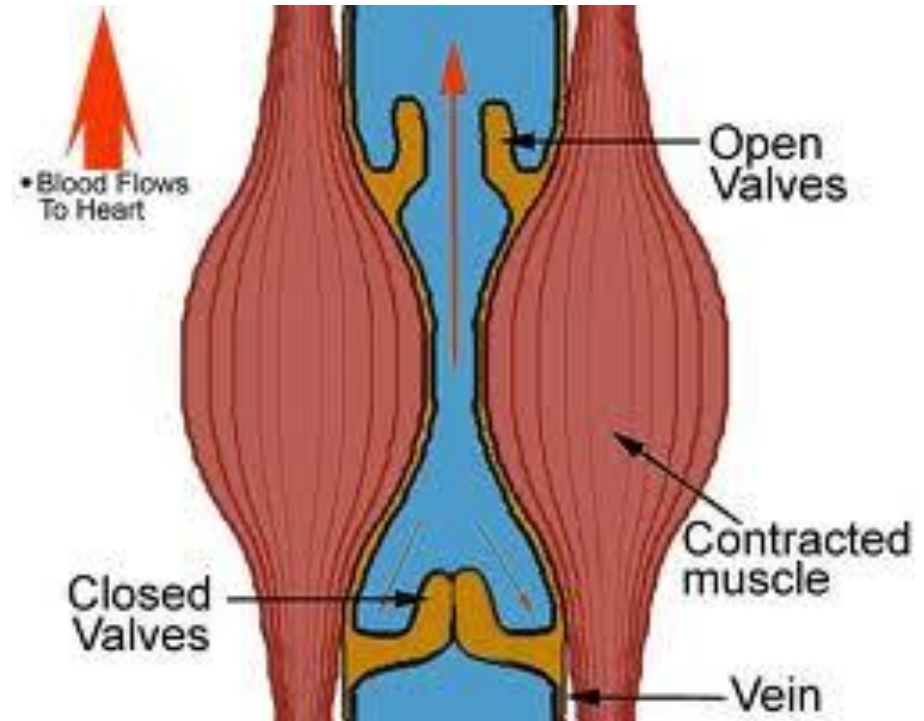
- Capillaries have very thin walls – a single layer of cells
- They allow for the interchange of gases, nutrients and waste products



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Veins - Venules

- Veins carry blood back to the heart
- Some veins have valves to prevent back-flow of blood



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THE HEART

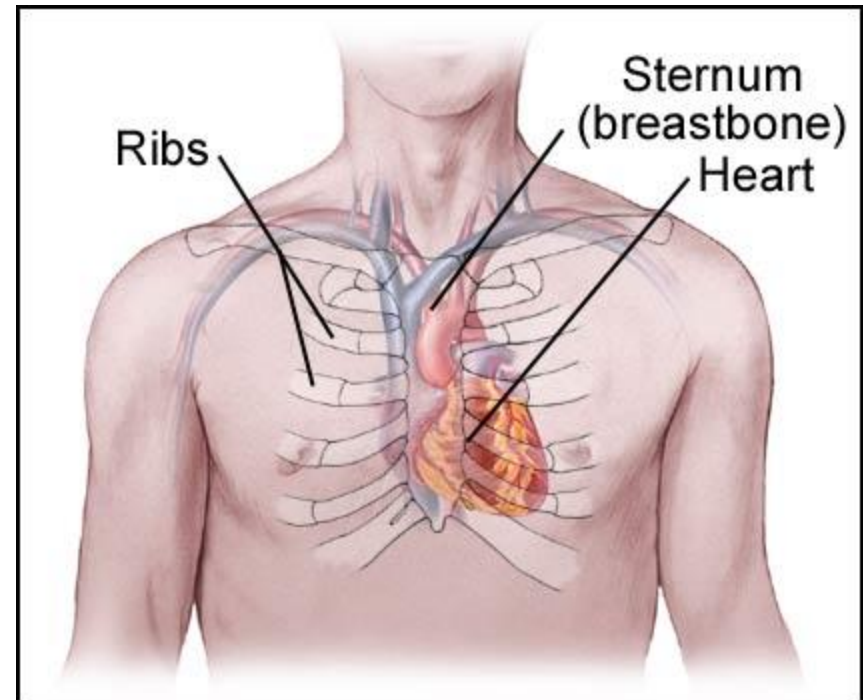


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Location of the Heart

Situated in the mediastinum
in the thoracic cavity

- behind the sternum,
- in front of the spine,
- between the lungs
- To the left of the midline

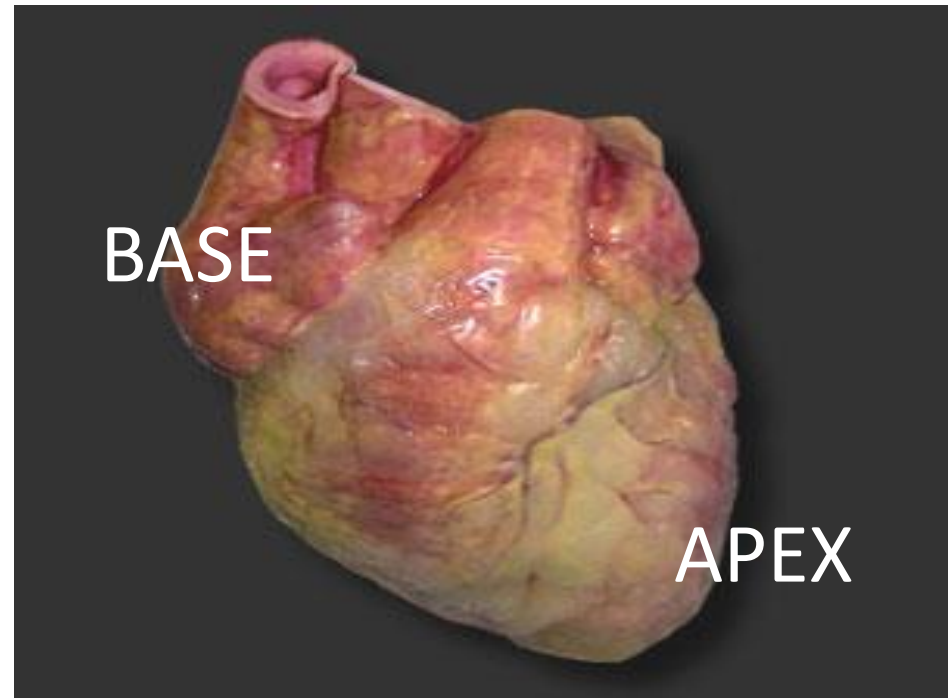


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It is described as having
an Apex and a Base

Cone shaped

Approximately the size of
the owners clenched fist

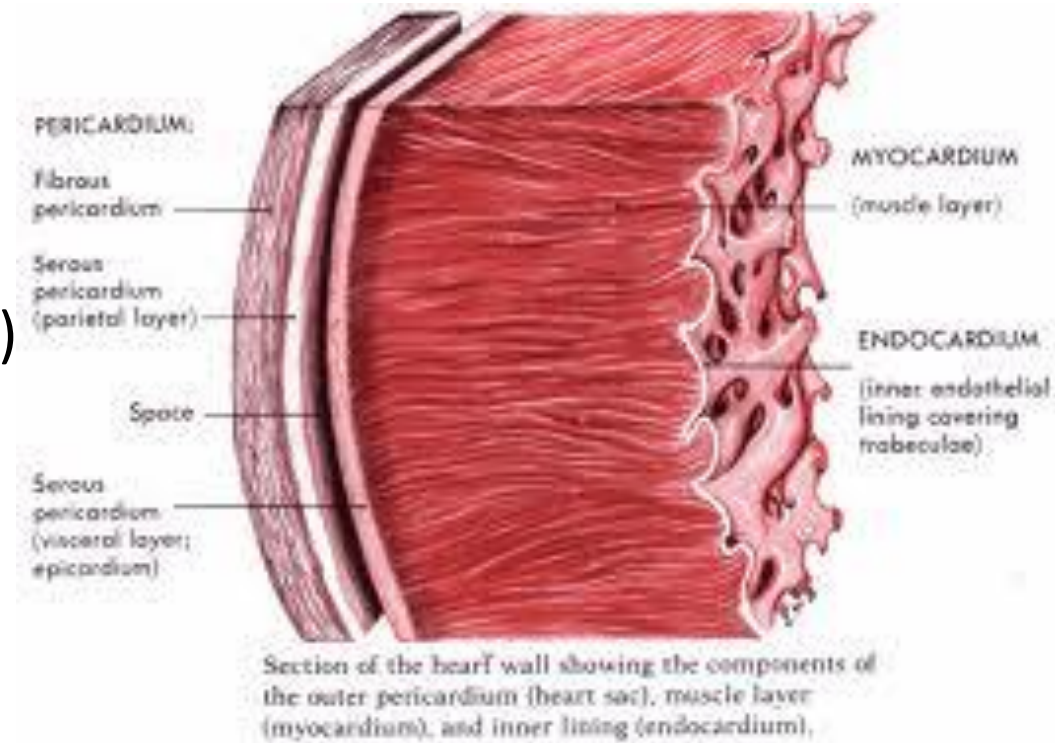


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Structure of the Heart

- The walls of the heart are composed of three main layers:

- Pericardium (outer layer)
- Myocardium (middle layer)
- Endocardium (inner layer)



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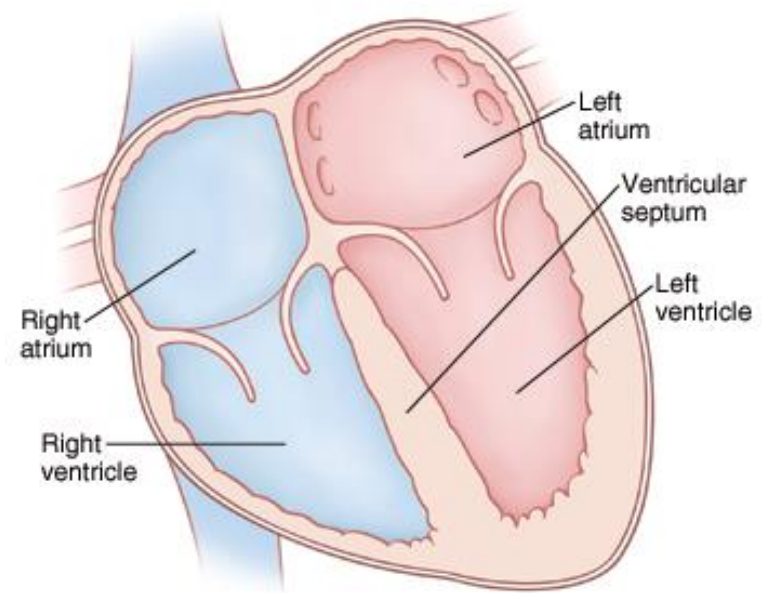
Structure of the Heart

The heart is comprised of four chambers:

Left and right atria

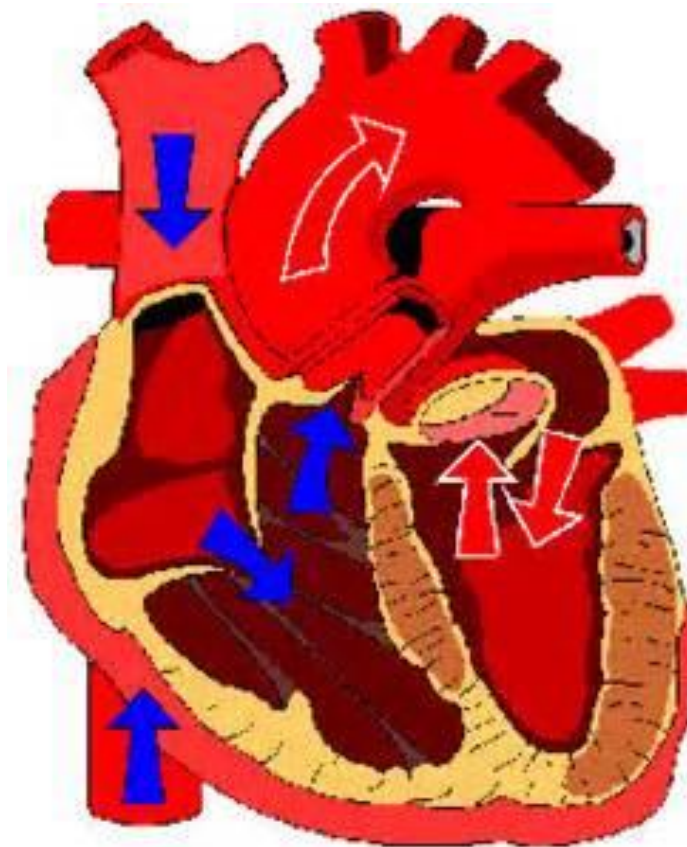
Left and right ventricles

All separated by valves



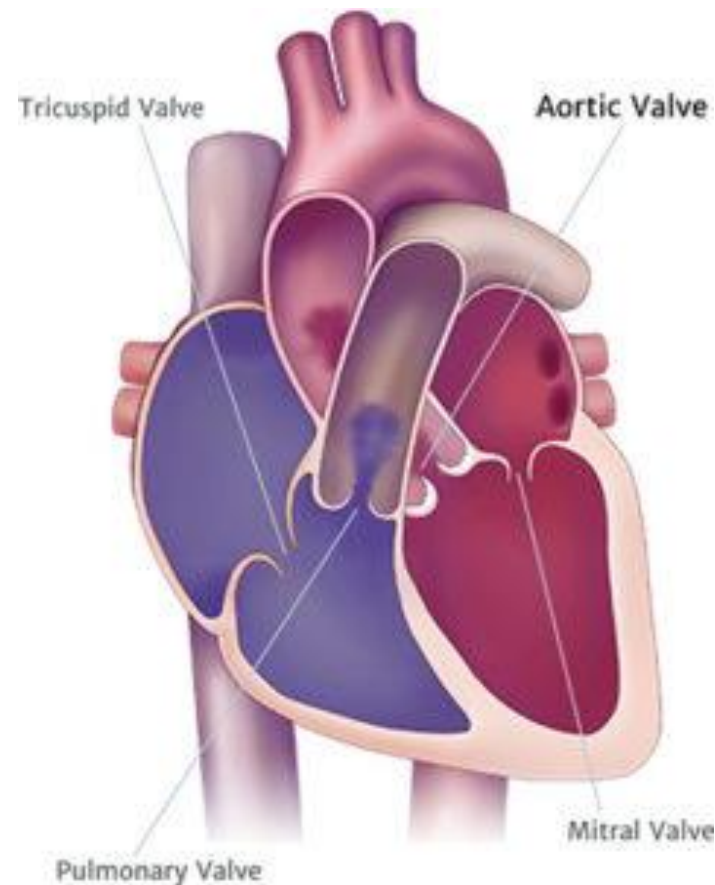
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Blood flow through the Heart



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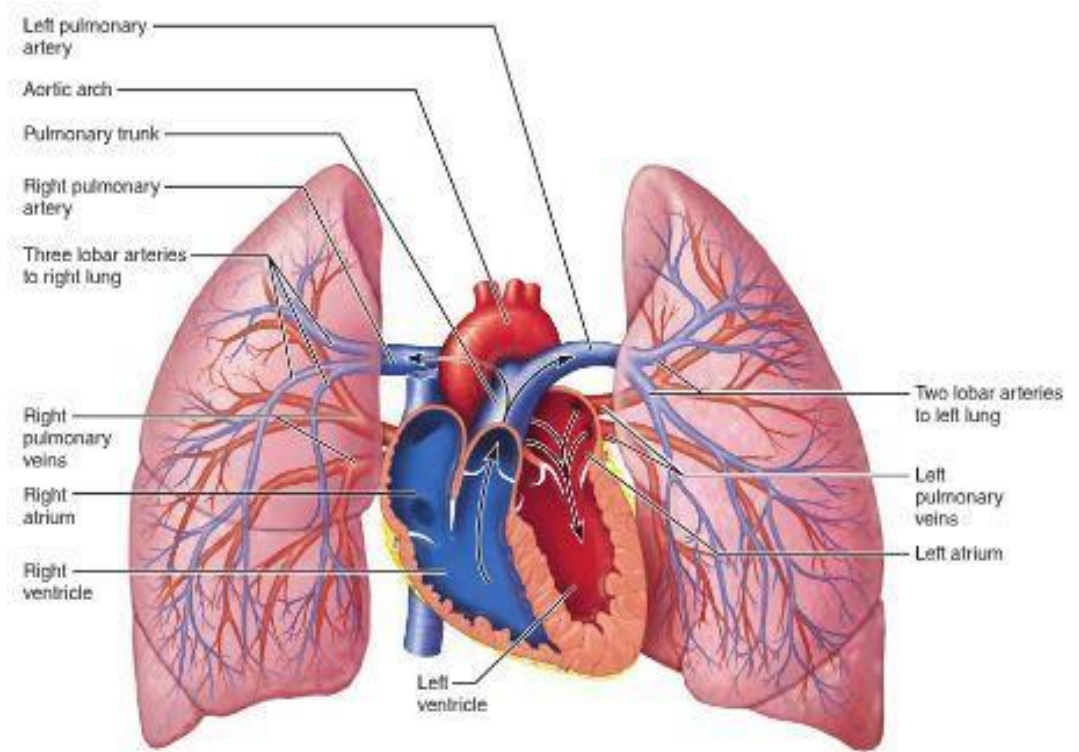
Heart Valves



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Pulmonary Circulation:

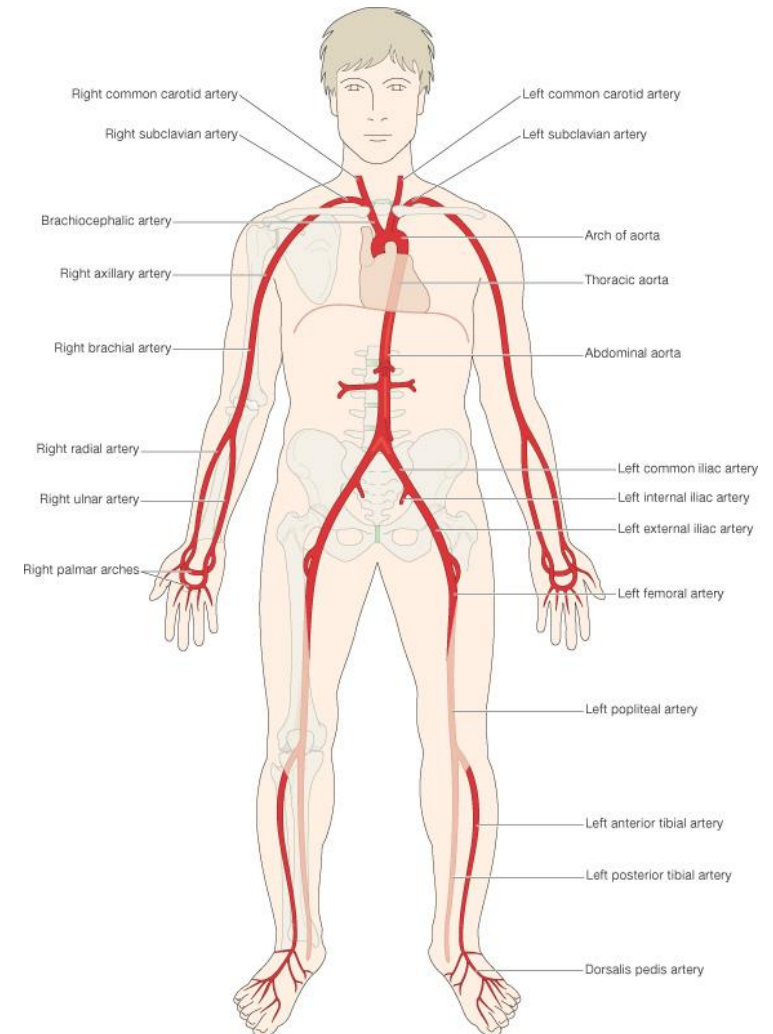
Deals with the passage of blood through the lungs



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Systemic Circulation:

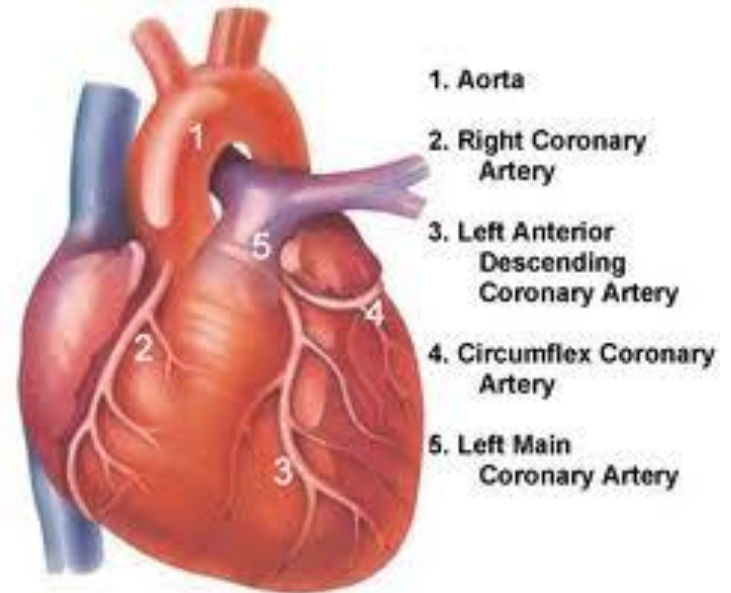
Deals with the passage of blood around the body



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Coronary Circulation:

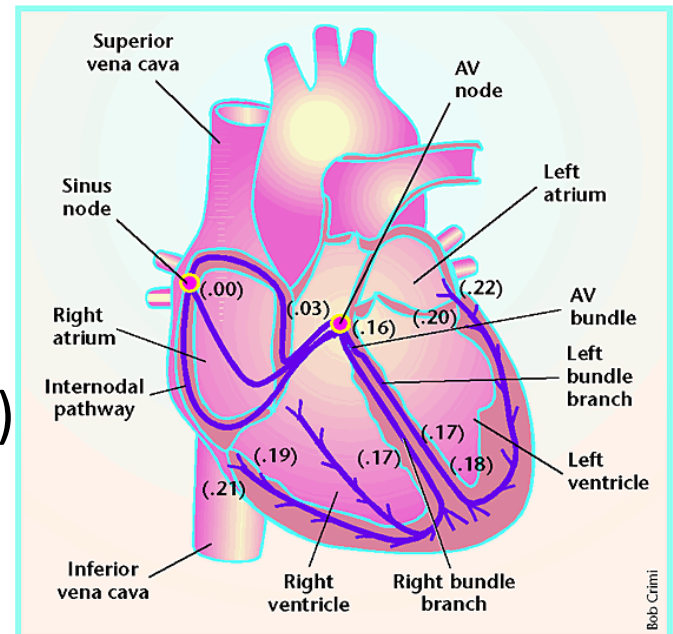
Deals with the supply of blood to the heart muscle



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Electrical Conduction System

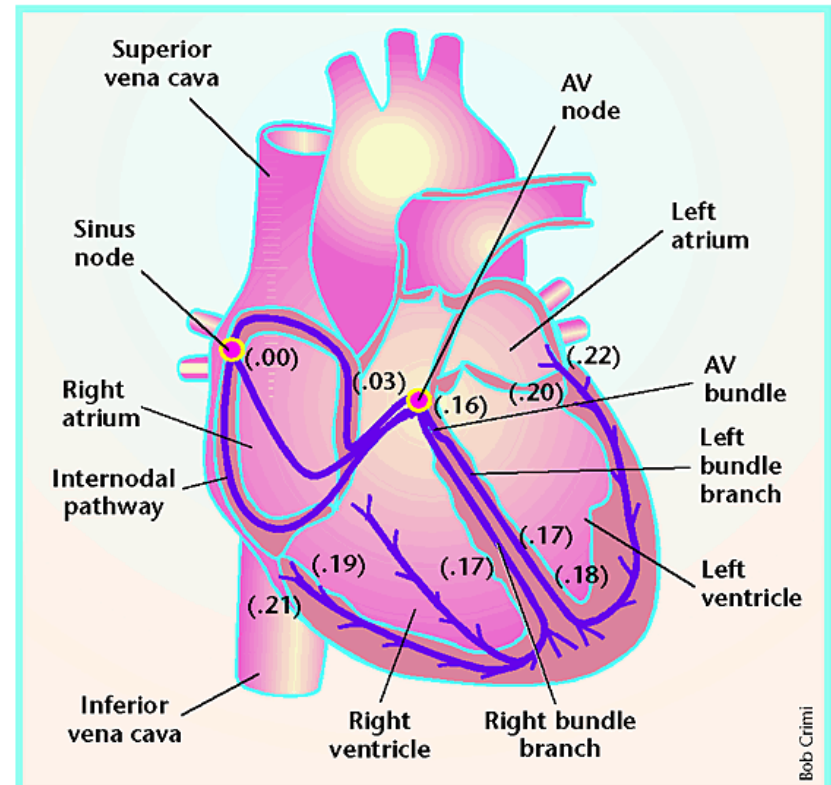
- Sino-Atrial node (SA)
- Atrio-ventricular node (AV)
- Bundle of His (Left and Right Branches)



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Sino-Atrial Node

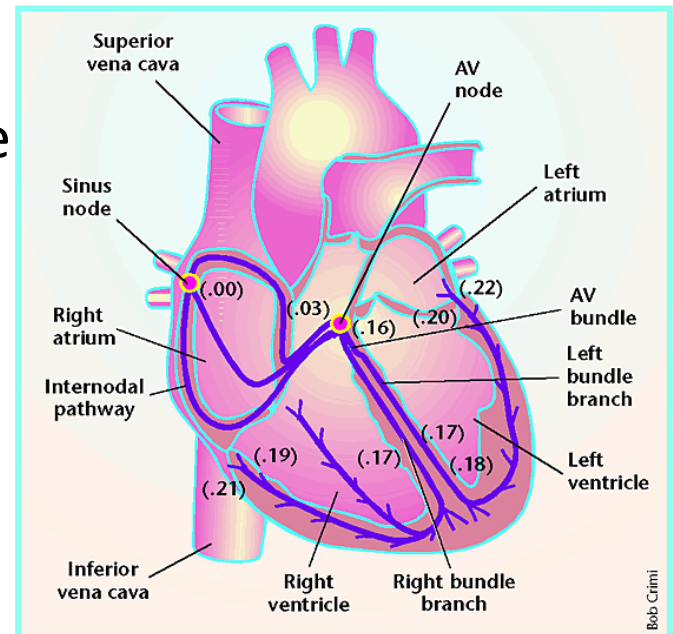
- Main pacemaker of the heart
- Found in right atrium
- Starts the electrical impulse
- Electrical impulses travel from SA node to AV node



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Atrioventricular Node

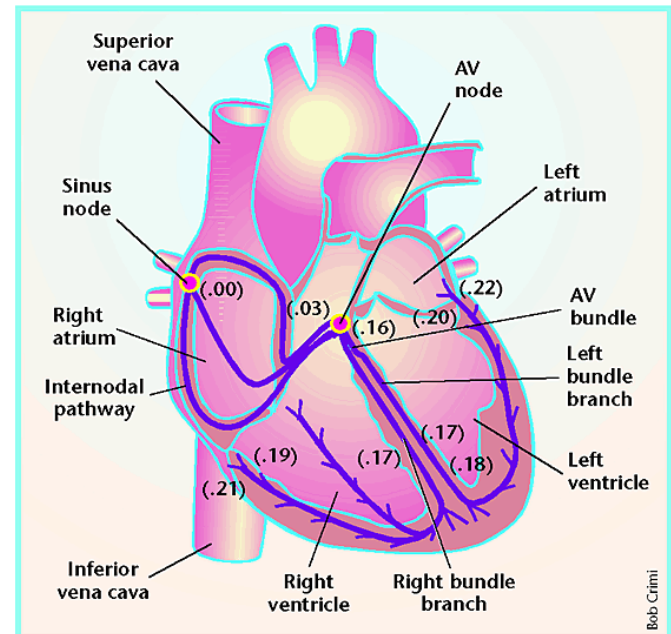
- Found in right atrial wall
- Delays electrical impulse from SA node to allow atria to fully contract
- Electrical impulse then released to the Bundle of His



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Bundle of His and Bundle Branches

- Bundle of His takes the impulse and splits into left and right bundle branches
- Left bundle transmits the impulse to the left ventricle
- Right bundle transmits the impulse to the right ventricle



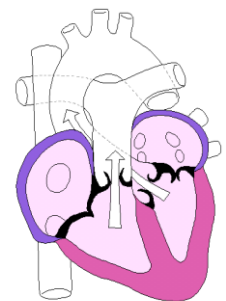
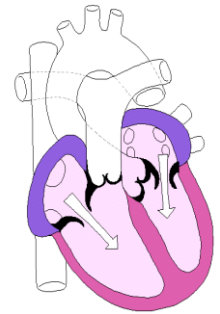
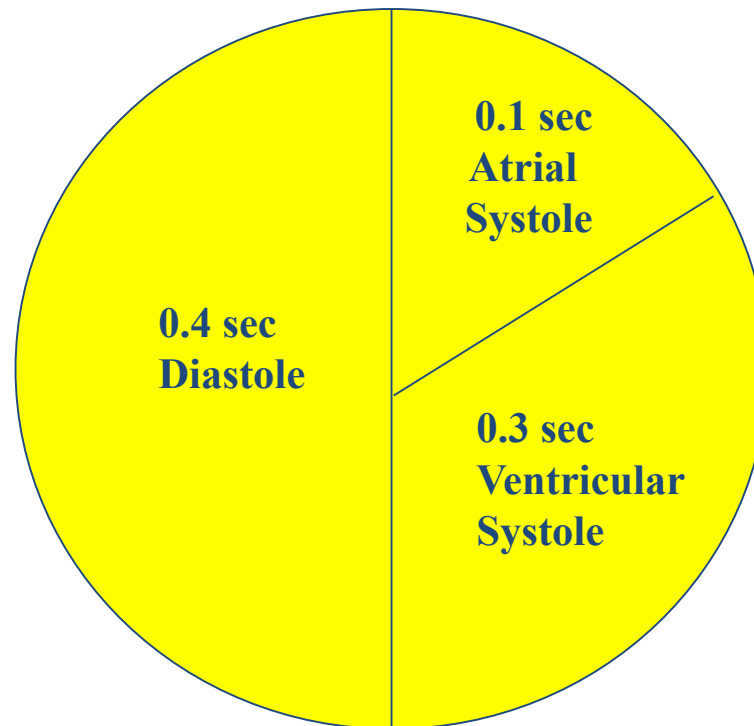
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The Cardiac Cycle

Atrial systole –
contraction of atria

Ventricular systole –
contraction of ventricles

Cardiac diastole –
relaxation phase



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Any Questions?